

1. Explain Docker layered architecture with diagram. What are the advantages of using container over virtual machine in a cloud environment?
2. What is continuous integration? Explain each stage of the software development life cycle of continuous integration with an appropriate diagram.
3. Explain the following in Configuration Management:
 - a. IaC
 - b. What are the two approaches to IaC
 - c. What are the two methods of IaC
 - d. Configuration Drift
4. Explain the comparison between traditional, Agile and DevOps with suitable diagrams. What are the technical and business benefits of DevOps?
5. Explain the following with a suitable diagram
 - a. Distributed version control system
 - b. git work flow
 - c. git clone
 - d. git fork

MCQs

1. Which is the right way to map a port from container to a port on machine which is running docker engine?
 - a. <container-port>::<local-port>
 - b. <container-port>:<local-port>
 - c. <local-port>:<container-port>
 - d. <local-port>::<container-port>
2. When a fork is done on a repository, which of the following is true?
 - a. The fork helps to add the user as a new contributor in the repository being forked.
 - b. The entire repository is copied and creates a new repository in the user's account.
 - c. The user is not allowed to commit in his forked repository without permission from the account holder of the repository being forked.

d. The user is allowed to commit in the repository being forked without any permission from the account holder.

3. What are the key components of DevOps?

- a. Continuous Integration
- b. Continuous Testing
- c. Continuous Delivery & Monitoring
- d. All of the above

4. Which of the following is correct with respect to merging two branches in git?

- a. The first branch is merged with second branch with the commit of the first branch as last commit.
- b. The first branch is merged with second branch with the commit of the second branch as last commit.
- c. If both the branches are not conflicting, a fast forward merge is created, with a new merge commit as last commit in the history.
- d. There is no way to merge branches with conflicts.

5. Which DevOps principle appreciates measuring processes, people, and tools?

- a. Continuous improvement
- b. Create with the end in mind
- c. Cross-functional autonomous teams
- d. People responsibility

6. In order to minimize risks to the business within DevOps, what is a main goal of version control?

- a. Ensure the ability to share the source code between different developer's teams
- b. Ensure the ability to re-create the previous state of the test environment
- c. Ensure the ability to alert when configuration changes from the desired state
- d. Ensure the ability to re-create the production environment and build processes

7. Which tool can be used to implement IaC?

- a. Chef
- b. Ansible
- c. Puppet
- d. all of the options

8. What's the correct sequence of continuous delivery workflow?

- a. git clone, compile, package, deploy, unit test
- b. git clone, compile, unit test, package, deploy
- c. package, git clone, unit test, compile, deploy
- d. git clone, unit test, compile, package, deploy

9. Which network protocol does Ansible use by default to communicate with managed nodes?

- a. HTTP
- b. HTTPS
- c. SNMP
- d. SSH

10. Which of the following team gets priority in DevOps?

- a. Operational Team
- b. Development Team
- c. Both Operational and Development teams
- d. None of the above

11. What is a Dockerfile?

- a. A template used to describe the build of an image
- b. Dockerfile keeps docker logs
- c. A folder or document to run Docker components
- d. A report or document that Docker components produce

12. Which of the following statement best describes the goal of DevOps?

- a. Establish an environment to release more reliable applications faster
- b. Establish an environment where the release of applications is valued more than its quality
- c. Establish an environment where application development performs all the operation tasks

d. Establish an environment where change management does not control application releases

13. What is NOT a challenges between the Development and Operations teams in a traditional organizations?

- a. Blame game between Dev and Ops
- b. Different tools used between Dev and Ops
- c. No feedback loop between Dev and Ops
- d. Development and Operations is not maintained by the same person

14. Docker attach is a command used to:

- a. Attach the running container to TTY.
- b. Attach the volume to a container.
- c. Attach the built image to TTY.
- d. None of the above

15. Which of the following terms best describes the Ansible architecture?

- a. Event-driven
- b. Client/Server
- c. Agentless
- d. Stateless

Answers -

1.c 2.b 3.d 4.c 5.a 6.d 7.d 8.b 9.d 10.c 11.a
12.a 13.d 14.a 15.c